Statistical Evaluation of AI

Carnegie Mellon University · Statistics & Data Science · Spring 2027

Course information

Course number	TBD; cross-listed for undergraduate and master's students
Units	9 (anticipated)
Meeting days and time	Tuesdays and Thursdays, 80 minutes; time TBD
Location	TBD
Course website	Canvas, link TBD
Final examination	No traditional final exam; the final project, presentation, and oral defense form the culminating assessment

Course staff

Instructor

Shannon Gallagher
Email: TBD
Office: TBD
Office hours: TBD and by appointment

Teaching assistants

Names, contact information, office hours, and responsibilities: TBD.

Questions whose answers may help the entire class should be posted to the designated Canvas discussion board. Personal matters and questions involving grades should be sent privately. Course staff will ordinarily respond within two business days; students should not expect immediate responses at night or on weekends.

Course description

Modern AI systems arrive with benchmark scores, probability claims, and confident recommendations. This course asks what those numbers actually mean.

Students will study AI systems through a statistician's perspective. They will define populations and estimands, investigate how data and labels were produced, assess the quality and assumptions of benchmarks, quantify uncertainty, compare systems fairly, test consequential assumptions, and interpret results in context.

This class **is not** a survey of machine-learning algorithms but rather a study of how a statistician or data scientist should use these systems. In this course, students will both use statistical thinking to evaluate AI systems and

use AI to enhance their statistical output, thus creating a feedback loop where one strategy can propel the other forward.

Course goals

The course has two linked goals:

1. Develop statisticians who can evaluate empirical claims about AI systems.
2. Develop statisticians who can use AI tools productively without outsourcing statistical judgment.

Every substantial analysis follows the same five-stage workflow:

1. Define the research question, hypotheses, and estimands.
2. Explore the data and assess its quality.
3. Model the data, make inferences, and evaluate performance.
4. Test how sensitive the conclusions are to key assumptions.
5. Make a decision and state its limitations and next steps.

Learning objectives

By the end of the course, students will be able to:

1. Translate a claim about an AI system into a statistical question with a target population, unit, and estimand.
2. Evaluate how data and labels were produced and how that shapes what a benchmark measures.
3. Quantify uncertainty in AI performance and compare systems fairly against a statistical baseline.
4. Judge whether a system's probability claims are calibrated and whether its performance carries over to new populations.
5. Use AI tools in statistical work while checking and documenting what they contribute.
6. Make and defend a recommendation whose strength matches the evidence.

Audience and prerequisites

The course is for advanced undergraduates and master's students. The prerequisite is 36-401 (Modern Regression) or an equivalent course. Students should be comfortable with:

- interpreting regression coefficients and fitted probabilities;
- confidence intervals and hypothesis tests;
- basic data manipulation, visualization, and programming;
- explaining a statistical result in writing.

Further coursework in statistics or machine learning helps but is not required. Machine learning is not the point of this course. We study AI systems only as far as we need to evaluate them, and Week 1 builds up to neural networks starting from regression.

Course format

Weeks 1–7

Tuesdays usually introduce a statistical problem and work through examples. Thursdays usually pair a shorter lecture with a hands-on lab. A short online checkpoint quiz closes out each week.

Weeks 9–15

After spring break, the course shifts to the team project. One class each week introduces a new statistical topic. The other is mostly a project work session, where teams work together in class.

Final project

The final project is the center of the course. Students work in teams of four as a research team for a client. The client has an AI system, some data, and a decision to make, and the team designs and carries out a statistical study to inform that decision. Every team fits a statistical baseline it can explain, compares it fairly with the AI system, quantifies uncertainty, tests whether its conclusion survives a reasonable alternative analysis, and makes a recommendation that matches the evidence.

Teams choose one of four tracks: food tagging, pet tagging, human preferences, or instruction following. Data, cached AI outputs, and starter code are provided for every track. The study is built in three milestones and presented to the client as slides and a five-minute video, followed by an oral defense.

Two documents on the course website's project page hold every project requirement: the **Project Description** (what to do and when) and the **Project Rubric** (how it is graded, in points).

Major topics

- From regression to neural networks
- Data and labels as measurements
- Evaluation design, including estimands, splits, and leakage
- Uncertainty in benchmark performance
- Comparing systems on the same cases
- Calibration and decision rules
- Distribution shift and sensitivity analysis
- Designed evaluations and LLM judges

Data and computing

Labs and homework use a few datasets that recur through the semester, plus others when a topic calls for it. The final project uses the track datasets described above.

We use Python in Google Colab on free, CPU-only resources. Data subsets, starter code, and cached AI-system outputs are provided, so no assignment requires a paid service, a GPU, or a large download. Rules for shared lab notebooks, service outages, and typed versus handwritten work are on the course technology page in Canvas.

Do not upload course data or project materials to an outside service, including an AI tool, unless the assignment says you may.

AI access

We assume you have access to at least one general-purpose AI assistant. You do not need to pay for one. Free options include:

- Microsoft Copilot Chat and Google Gemini and NotebookLM, available at no cost with your CMU account. Sign in with your Andrew ID to get CMU's data protections.
- The Gemini assistant built into Google Colab.
- The free tiers of ChatGPT and Claude.

No assignment requires a paid plan. Free tiers have usage limits, so don't leave AI-assisted work until the last hour.

Readings and learning resources

Textbook. Christopher M. Bishop and Hugh Bishop, *Deep Learning: Foundations and Concepts* (2024). A free online version is available at bishopbook.com. We use it as a technical reference for the AI systems we study.

I will link short readings and sources as we go.

Assessment and grades

Component	Weight	Purpose
Homework (seven, lowest dropped)	15%	Practice with feedback; AI allowed where each assignment says
Weekly checkpoints (ten, lowest two dropped)	20%	Short online quizzes on each week's lectures, without AI
In-class midterm	25%	A challenging exam on the first half of the course, without AI
Final project, including peer assessment	40%	Team statistical evaluation study, video with slides, and oral defense

Homework is where you practice, with AI assistance where permitted. Checkpoints, the midterm, and the oral defense are where you show, without assistance, that the reasoning is your own.

Letter grades

Letter grades use these thresholds. Master's students receive graduate letter grades, which include plus and minus grades.

Course percentage	Undergraduate	Master's
93–100	A	A
90–<93	A	A−
87–<90	B	B+
83–<87	B	B
80–<83	B	B−
77–<80	C	C+
73–<77	C	C
70–<73	C	C−
67–<70	D	D+
60–<67	D	D
<60	R	R

Your master's program sets its own minimum passing grade.

Both scales follow CMU's grading standards. Individual assignments may use points or rubric categories. Final course percentages are computed from the weights above; the course is not graded by ranking students against one another.

Homework

There are seven graded homeworks, and the lowest score is dropped. Homework is usually due Tuesday at 11:59 pm. Project milestones are separate from homework.

Each homework has three problems. Problem 1 is a derivation or hand calculation done without AI, and it is good practice for the midterm. Problem 2 is a data analysis that picks up where the lab left off. Problem 3 asks you to use an AI tool and then check and correct what it produced.

You may reuse code from lab. The analysis and the writing you turn in must be your own.

You do all homework in a Colab notebook that we provide. Submit two files to Gradescope, a PDF of the finished notebook and the notebook itself. Handwritten work is welcome but never required. You can type math in the notebook, or insert a photo of handwritten work followed by a typed sentence stating your conclusion.

Before HW1, you will complete HW0. It is an ungraded practice run of the whole process, so submission problems get fixed before anything counts.

Labs and project work sessions

A lab is a guided coding session in class. The instructor works through an example in a Colab notebook. Then students change the code, run it, and talk through the results with classmates. Labs vary from week to week, but most are set up so that the homework starts where the lab leaves off and reuses its code.

Labs are not graded. Skipping one will usually make that week's homework harder.

After spring break, Thursdays are mostly project work sessions. Teams work on their projects in class, often after a short guided exercise that builds part of the next milestone. This work is not graded separately.

Weekly checkpoints

Each week ends with a short online checkpoint quiz in Canvas. It opens after Thursday's class, covers that week's lectures and lab, and is due the following Tuesday at 11:59 pm. Once you open it, you have 20 minutes to finish, and you get one attempt.

Checkpoints are individual. You may use your own notes, but not AI tools, classmates, or any other outside help. Questions come directly from that week's lectures, so if you followed lecture and reviewed your notes, a checkpoint should not take long. They are also good practice for the midterm.

There are ten checkpoints, one for each of Weeks 1–6 and 9–12. Your lowest two scores are dropped. That covers ordinary illness, travel, and busy weeks, so there are no makeups or extensions. If something serious affects more than two checkpoints, contact the instructor.

If attendance starts to drop, I will move checkpoints into class time.

Midterm

The midterm is in class on Thursday, March 4, and is designed to be challenging. It is completed individually without AI tools, networked devices, or books. You may bring **one sheet of notes you prepared yourself, front and back** (US letter or A4, handwritten or typed). There is no final exam.

Final project

Each team carries out a full statistical evaluation of one AI system for a client, following the five steps above. Every project includes a simple reference point (such as always predicting the most common class) and a statistical baseline the team fits and can explain, usually logistic regression. It also includes uncertainty for each key estimate, a fair comparison between the baseline and the AI system, one sensitivity analysis that could change the conclusion, and a recommendation that matches the evidence.

The project is 40% of the course grade, including an individual peer assessment. The Project Rubric lists every deliverable and its points, and the Project Description says what each one contains. Both are on the project page.

Course calendar

Dates follow CMU's official Spring 2027 academic calendar. Homework is due Tuesdays at 11:59 pm unless noted. Each week's checkpoint opens after Thursday's class and is due the following Tuesday at 11:59 pm. Canvas contains the authoritative dated calendar.

Week	Tue / Thu	Topic	Checkpoint	In class and due
1	Jan 19 / 21	From regression to neural networks	CP1	Baseline lab · HW0 due Thu Jan 21
2	Jan 26 / 28	Data as measurement: labels and repeated judgments	CP2	Label-measurement lab · HW1 due Jan 26
3	Feb 2 / 4	Evaluation design: estimands, splits, and AI-assisted workflow	CP3	Evaluation-design lab · HW2 due Feb 2
4	Feb 9 / 11	Benchmark performance as estimation	CP4	Interval lab · HW3 due Feb 9
5	Feb 16 / 18	Paired system comparison	CP5	Paired-comparison lab · HW4 due Feb 16
6	Feb 23 / 25	Calibration and decision-rule sensitivity	CP6	Calibration lab · HW5 due Feb 23
7	Mar 2 / 4	First-half synthesis	—	Review Tue · **in-class midterm Thu Mar 4** · project teams finalized
8	Mar 9 / 11	**Spring break**	—	No class or assignment
9	Mar 16 / 18	Distribution shift and project launch	CP7	Shift exercise · project launch session Thu
10	Mar 23 / 25	Sensitivity analysis	CP8	Project Milestone 1 due Mar 23 · HW6 due Mar 30
11	Mar 30 / Apr 1	Designed AI evaluations	CP9	Project work session Thu
12	Apr 6 / 8	LLM judges as measurement instruments	CP10	Project Milestone 2 due Apr 6 · project work session Thu · HW7 due Apr 13
13	Apr 13 / —	Human-centered and responsible AI	—	Guest lecture: Carol J. Smith (Tue) · no class Thu (Spring Carnival)
14	Apr 20 / 22	Statistical argument and recommendation	—	Storyboard in class Tue · Project Milestone 3 due Apr 20 · dry run Thu
15	Apr 27 / 29	Oral defenses and course synthesis	—	Slides and video due Sun Apr 25 · oral defenses Tue and Thu

Registrar dates. The semester drop deadline is Mon Mar 1, and by then you will have feedback on several homeworks and checkpoints. The pass/no pass and withdrawal deadline is Mon Apr 5. The last day of classes is Fri Apr 30, and there is no final exam during the May 3–11 exam period.

The instructor may adjust the pace or readings in response to student learning and developments in AI evaluation. Changes to graded work or deadlines will be announced on Canvas with reasonable notice.

Course policies

Attendance and participation

Regular attendance is strongly encouraged. Labs and project work sessions are hard to make up from slides alone, and they are where homework and project work usually start. Attendance is not graded. If attendance drops, the weekly checkpoint quizzes will move into class time.

If illness, a religious observance, university travel, disability-related needs, or an emergency prevents attendance, notify course staff as soon as reasonably possible. Students are not expected to disclose private medical details. We will identify an appropriate way to recover the essential learning or artifact.

Late work

There are no flex days. If you need more time on an assignment, ask the instructor before the deadline. Extensions of up to 24 hours are considered case by case.

Weekly checkpoints do not get extensions, because the lowest two scores are already dropped. University-approved accommodations and serious emergencies are handled individually.

Work submitted late without an approved extension may receive reduced or no credit.

Collaboration

Statistical work benefits from discussion, but collaboration must remain visible and must not replace individual understanding.

- **Labs.** Work with whomever you like. Talking through the lab with classmates is encouraged.
- **Homework.** You may discuss concepts, debugging, and the lab code with classmates. You must run the analysis, make the decisions, and write the code and prose yourself. Do not share finished answers, solution files, or text or code that others could copy.
- **Weekly checkpoints.** These are individual. Your own notes are allowed, but not AI tools, classmates, or other outside help.
- **Midterm.** No collaboration or outside help.
- **Project.** Collaborate within your team. Sharing project work outside your team requires instructor permission.

Every individual submission must list collaborators and other assistance, or state that none was received. Every external source, including borrowed code and AI assistance, must be acknowledged where required.

AI tools

AI is part of how this course is taught, and two rules apply whenever you use it. You are responsible for every statement, number, citation, and line of code you turn in. You must also record what the AI contributed and how you checked it, so another analyst could follow what happened.

AI is allowed, and sometimes required, on homework and the project. Each assignment says how. Sometimes you will use AI to explain an idea or draft code, and sometimes the AI system is the thing you are evaluating. Making up citations, turning in output you didn't check, or passing off AI work as your own is not acceptable.

AI is not allowed on weekly checkpoints or the midterm. "The AI said so" is never statistical evidence.

AI built into other tools counts too. That includes the Colab assistant, chat panels in editors, and autocomplete that writes code or prose for you. Turn these off when an assignment or problem says no AI. Don't give them course data unless the assignment says you may.

Academic integrity

Honesty, attribution, and transparency are part of the statistical practice this course teaches. Students must comply with Carnegie Mellon's Policy on Academic Integrity. The policy requires original work and acknowledgement of permitted collaboration and assistance. In this course, that acknowledgement includes people, sources, software, and AI tools as described above.

Copying another student's work, providing copyable individual solutions, using unauthorized assistance, fabricating data or citations, misrepresenting an AI-generated contribution, or using prohibited tools on an assessment may constitute an academic-integrity violation. Ask before submitting if the boundary is unclear.

Regrade requests

Requests to correct a grading or scoring error must be submitted in writing within seven calendar days of the graded work being returned. The request should identify the relevant rubric criterion and explain the suspected error. A regrade reviews the relevant work for accuracy and consistency; it is not a negotiation for additional points. Clerical errors should be reported at any time.

Devices and recording

You will need a laptop for labs and some class activities. During discussion, I may ask you to close it.

Do not record class (audio, video, photos, or screenshots) without my written permission or an approved disability accommodation. Any approved recording is for your own use and may not be shared. This protects class discussion, student work, and guest speakers.

If course technology goes down

If Colab, Canvas, or another course service goes down, save what you can, note when it happened, and keep working on anything that doesn't need it. Don't pay for a substitute or move course data somewhere else. If a service is down for more than 15 minutes during class, we will finish the lab another time. If a confirmed outage hits in the last six hours before a homework or project deadline, everyone gets an extension, announced on Canvas. If your own computer or internet fails, tell us right away and we will help you find another way to work or submit.

Respect, inclusion, and religious observances

Different experiences and perspectives are an asset in a course about measurement, populations, human judgments, and consequential decisions. We will critique claims and methods rigorously while treating classmates and the people represented in data with care. Students should use one another's stated names and pronouns and make room for equitable participation in group work and discussion.

Students who experience a conflict with a religious or cultural observance should contact the instructor as early as practical. The course calendar will be checked against CMU's inclusive holiday and interfaith calendar when dates are finalized.

Accommodations for students with disabilities

If you require accommodations that are not yet approved, apply through the Disability Resources Online Portal. If accommodations are already approved, use the portal's Accommodations Management section to notify the instructor and discuss your needs as early as possible. The instructor will work with you and the Office of Disability Resources to provide appropriate accommodations.

Mandatory reporting

CMU employees, including instructors and teaching assistants acting in their course roles, are required to report information about possible discrimination or sexual misconduct to the Office for Institutional Equity and Title IX. That office will offer information and support; the student decides whether and how to respond. Students who prefer to speak with someone who is not required to report can use CMU's confidential reporting resources.

Student well-being and campus resources

Academic stress and difficult life events can interfere with learning. Asking for support early is a sound decision, not a failure. Students are welcome to contact the instructor when course circumstances are contributing to a problem; private details are not required to ask for help.

- Counseling and Psychological Services (CaPS) provides confidential support. Call **412-268-2922** for assistance; phone support is available 24/7.
- For an imminent, life-threatening emergency, call **CMU Police at 412-268-2323** or **911**.
- Student Academic Success Center services may be used consistent with the acknowledgement rules above.
- CMU Pantry and other well-being resources are available to undergraduate and graduate students.

How to succeed in this course

- Treat every headline number as an estimate. Name its target population and uncertainty.
- Keep a running analysis notebook and AI-use log rather than reconstructing decisions at the deadline.
- Finish each lab. It usually sets up the homework.
- Review your lecture notes before each checkpoint, and do homework Problem 1 without AI. Both prepare you for the midterm.
- Read for a purpose. Identify the figure, assumption, or documentation detail the week's analysis uses.
- Start project questions with the client decision, not with an available metric.
- Use office hours before a small design ambiguity becomes a large coding problem.
- Be prepared to explain any submitted result without relying on the AI tool, teammate, or code that produced it.

Syllabus changes

This syllabus describes the intended course. The instructor may make reasonable changes to topics, readings, examples, or dates in response to the official university calendar, guest-speaker availability, or student learning. Material changes will be announced through Canvas, and the dated Canvas schedule will control when it differs from this planning document.